

Reinforcement Learning-Enabled Terrain Adaptation for Bipedal Wheeled-Legged Robots with Height Adjustment Capability

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Abstract—For the bipedal wheeled-legged robot (B-WLR), maintaining posture adjustment while adapting to complex terrains remains a key challenge to its broader deployment. To overcome this challenge, this study presents a reinforcement learning-based control framework that integrates terrain adaptability with posture regulation ability. By defining a mapping between the commanded body height and joint angles, also incorporating an active linear interpolation method, the robot gains the capability to adjust its height across varying terrains while maintaining stability. Meanwhile, to ensure efficient terrain traversal, an active evaluation mechanism is designed based on discrepancies among body velocity, commanded forward velocity, and wheel speed. Through this mechanism, the B-WLR lifts its legs when encountering vertical obstacles while relying on wheeled locomotion on other terrains, thereby enhancing adaptability to complex environments. The proposed strategy is validated through Webots simulations and experimentally verified on the LimX Dynamics Tron1.

Index Terms—Wheel-legged robot, Reinforcement learning, Terrain adaptation, Motion control.

I. INTRODUCTION

The wheel-legged drive (WLD) architecture, which integrates the advantages of both wheeled and legged locomotion, enables efficient mobility in complex environments through adaptive gait transitions. Bipedal wheeled-legged robots (B-WLRs) exhibit superior flexibility and maneuverability compared with other forms of WLDs. However, this configuration faces significant challenges when deployed on terrains with pronounced elevation variations. Although B-WLRs perform reliably on flat surfaces, they often lack sufficient adaptability



Fig. 1. The first row, from left to right, presents the Webots simulations: adjusting height, climbing stairs, ascending slopes, and traversing rough terrain. The second row, from left to right, shows the corresponding experimental validations. Supplementary video can be found at: https://drive.google.com/file/d/1FsoPUZlac7x1-Ce8O_E8-E4OALCyg5IT/view?usp=sharing

to surmount unstructured terrains [1, 2]. In addition, most existing control frameworks depend on static gaits or predefined motion sequences, which further restrict their ability to cope with diverse terrain conditions [3].

On uneven surfaces, common approaches generally fail to correct the robot's posture or ground clearance in real-time,

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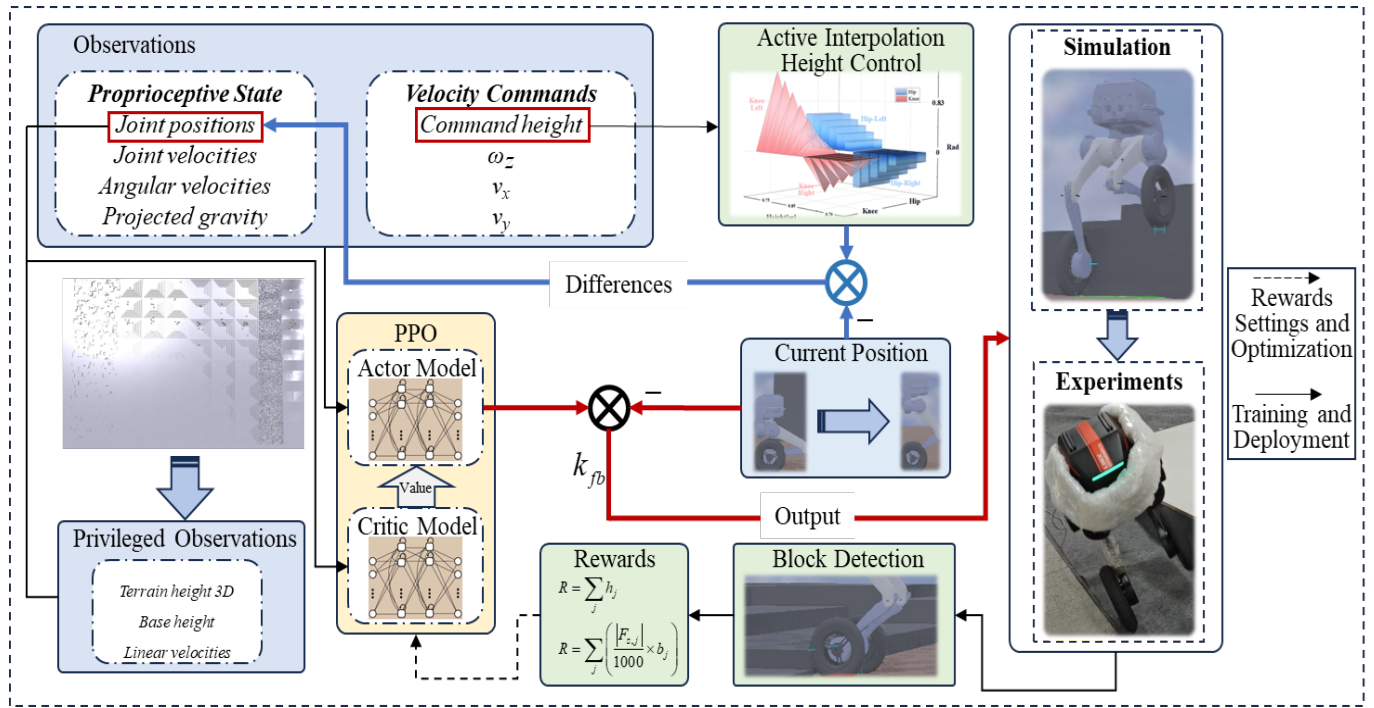


Fig. 2. Framework for the height control of a robotic system using PPO. The system receives proprioceptive states and privileged observations to generate velocity commands. The PPO model, consisting of an actor and a critic network, processes these inputs to provide active interpolation for height control. The system is validated in both simulation and real-world scenarios, with the performance evaluated in terms of rewards. Output: After optimization using the PPO algorithm, the difference between the position output by the Actor model and the current position is taken as the control output. Differences: The difference is defined as the deviation between the interpolated thigh and knee joint angles and those of the robot's current configuration.

thereby reducing its ability to traverse obstacles smoothly. Consequently, improving height adjustment capability while maintaining traversal capability has become a central issue in current research on wheeled-legged robots [4]. Recently, growing research efforts have focused on tackling the motion control challenges of B-WLRs in complex environments, with an emphasis on achieving active and precise body posture regulation while preserving efficient terrain traversal capability.

Conventionally, control methods for similar structures can be categorized into two parts [5]. One is the model-based control method, while the other is the data-driven approach, which has increasingly adopted reinforcement learning (RL) techniques in recent years [6].

In model-based control, Wensing *et al.* [7] provided a comprehensive review of model predictive control (MPC) and its applications in legged robots, noting that its performance is highly dependent on model accuracy. Bjelonic *et al.* [8] proposed a whole-body MPC framework combined with an online gait sequence generation method for wheeled-legged robots, enabling automatic transitions between wheel and gait patterns. To further enhance adaptability to unstructured terrains, Mason *et al.* [9] developed an MPC algorithm based on optimized contact forces, improving the robot's balance and driving capability on uneven surfaces. In conventional algorithms, precise posture regulation and multi-terrain traversal rely heavily on awareness of both the robot's internal states and external disturbances [10]. Agarwal *et al.* [11] introduced

a state estimation method based on the extended Kalman filter, laying the foundation for reliable motion control. To address disturbances caused by load variations, Liu *et al.* [12] designed a load perception and localization framework for heavy-duty quadruped robots without external force sensors. Minniti *et al.* [13] integrated an adaptive control Lyapunov function with MPC, allowing the robot to estimate and compensate for unknown load dynamics online while maintaining stable motion.

RL has demonstrated exceptional robustness in controlling complex dynamical systems, providing an effective paradigm for achieving both terrain adaptability and precise posture regulation in B-WLRs. Hwangbo *et al.* [14] demonstrated the capability of RL to acquire dynamic locomotion skills in legged robots with high efficiency. Rudin *et al.* [15] further advanced this by introducing large scale GPU-based parallel simulations, which improved training efficiency even more and allowed robots to acquire walking skills within minutes, achieving seamless zero-shot transfer to real-world platforms. Building on this foundation, they introduced an end-to-end task framework guided by positional objectives, endowing robots with autonomous path planning and adaptive gait generation for efficient obstacle traversal [16].

Lee *et al.* [17] employed RL to train a quadruped wheeled-legged robot, enabling online transitions between driving mode and walking mode in response to the user command and terrain. Chamorro *et al.* [18] proposed an RL framework for

B-WLRs that integrates positional objectives with Boolean terrain-switching logic, allowing the robot to autonomously engage a stair-climbing mode and ascend 15cm steps using only onboard perception. Li *et al.* [19] introduced a contact-triggered blind climbing framework, where wheel-obstacle contact forces initiate feedforward leg-lifting trajectories, allowing Tron1 to surmount obstacles exceeding their wheel-hub radius. Lee *et al.* [20] developed an integrated control framework combining RL, privileged learning, and hierarchical control, achieving seamless transitions between walking and driving modes in complex tasks.

Although the aforementioned studies have achieved promising results, existing control frameworks frequently focus primarily on static adjustment or emphasize terrain traversal performance alone, making it difficult to balance terrain adaptability and posture regulation [3], which constrained the B-WLR's adaptability in practical applications. This study proposes an RL-based control framework that integrates terrain traversal capability with active posture regulation. The main contributions are as follows:

- 1) A velocity-triggered RL framework is proposed, in which the obstacle detection is achieved through the computation of a tri-source velocity error(referred to Sec.II-D). By incorporating an active leg-lifting strategy, the B-WLR attains efficient terrain traversal capability across diverse environments.
- 2) A command height-joint angle mapping is established to maintain terrain traversal performance. By applying piecewise linear interpolation, the proposed framework enables rapid and adaptive posture adjustment.
- 3) The proposed RL framework is successfully deployed on the Tron1 robot. It demonstrates the ability to traverse 10cm stairs, a smooth slope, and dynamically perturbed inclines, while maintaining effective height regulation throughout these challenging terrains.

Fig.1 shows the simulation and experiments of the proposed framework. The rest of this paper is organized as follows: Sec.II shows the methodology, Sec III demonstrates the verifications, and Sec IV presents the conclusions.

II. METHODOLOGY

Fig.2 illustrates the RL framework used for training and deployment. In this subsection, we provide a description of the training learning environment, the RL task, the active interpolation height control, and the velocity-triggered strategy.

A. Learning Environment

1) *Simulator and Platform*: This study utilizes Isaac Gym for training and Webots for validation. During training, Isaac Gym's powerful parallel computing capabilities [21] enable the simultaneous execution of multiple environment instances on a single GPU, boosting policy learning efficiency. In the validation phase, Webots is used for high-fidelity simulation [22], offering a fast and efficient platform for verifying trained policies. Cross-validation is performed using the open-source Tron1 Unified Robot Description Format (URDF) model in

Webots [23], ensuring policy adaptability across various simulation environments. This process guarantees the reliable transfer of policies from simulation to real-world applications.

2) *Learning Algorithm*: The research employs a learning framework that combines Proximal Policy Optimization (PPO) with an asymmetric actor-critic architecture. Within this framework, the critic network performs value estimation by leveraging privileged information, then transmits the derived value to the actor network to guide its policy optimization. All training was conducted on a single NVIDIA RTX 4090 GPU, with convergence achieved within 6,000 iterations, demonstrating the algorithm's high efficiency.

B. Task Formulation

1) *State*: In this study, we adopt an asymmetric actor-critic framework, where the state is divided into observation and privileged information. The observation is available to both the actor and critic during training and deployment, while the privileged information is accessible only to the critic during training. Their detailed definitions are listed in Tables.I and II. The joint positions are the knee angle after active interpolation, which can be inferred from Fig.3.

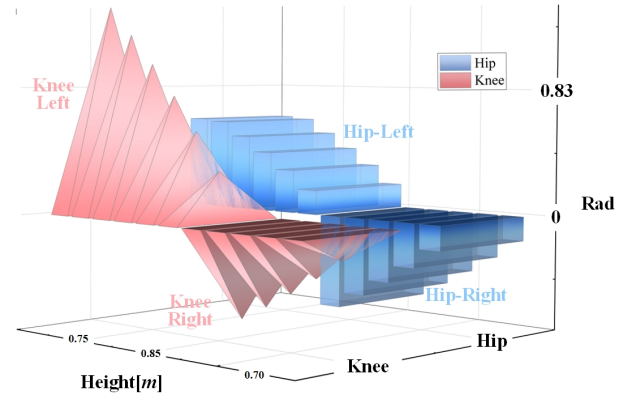


Fig. 3. Active interpolation data during the training, and the corresponding posture can be found in Fig.4

The aforementioned method effectively reflects the robot's relative position changes in the environment, particularly in complex or dynamic terrains. Through active linear interpolation control, the B-WLR can dynamically adjust according to predefined control objectives, ensuring that the height-joint position adapts to the real-world environment during task execution. Due to the difficulty of directly obtaining the robot's linear velocity and absolute base height of a B-WLR, this data is provided as privileged information and excluded from the actor network's observations. Additionally, the training environment supplies actual terrain height data to the privileged information, enabling the adjustment of relative height across various terrains.

2) *Actions*: In this study, the active numbers of joints of the tested robot are 8 (abads, hips, knees, and wheels), and the output dimension of the action was set to 8.

The actions command can be used as input to the PD controller, allowing the preset controller to adjust the joint status of the robot to achieve the target posture. As for the

TABLE I
OBSERVATION

Symbols	Name	Units	Coeff.	Size	Noise(%)
$\vec{q}$	Joint positions	rad	1.0	6	± 1
$\dot{\vec{q}}$	Joint velocities	rad/s	0.05	8	± 150
$\dot{\theta}$	Angular velocity	rad/s	0.25	3	± 20
$\vec{\gamma}$	Projected gravity	-	1.0	3	± 5
h_{target}	Height command	m	0.8	1	± 0

TABLE II
PRIVILEGED INFORMATION

Symbols	Name	Units	Coeff.	Size	Noise(%)
H_{terrain}	Terrain height	m	5.0	187	-
h	Base height	m	0.8	31	-
v_x	Linear velocity	m/s	2.0	3	-

wheel joint, it achieves the action control through the derivative control only, which means the position gains were set to zero during the motion control of the wheel. Additionally, we set the position gains for the joints and wheel individually to achieve the proper motion control.

3) **Rewards**: The main setting of the rewards is demonstrated through Table.III. The reward setting in this study was grant greater flexibility to traverse complex terrains, enabling solutions that are difficult for traditional algorithms to achieve. Reward #1 ensures the robot maintains a preset speed. Reward #2 promotes adaptive motion planning to avoid collisions in complex terrains. Reward #3 uses height and posture control to stabilize the B-WLR on uneven terrain. Reward #4 maintains reasonable joint torque and motion ranges during movement and height adjustment. Reward #5 improves training stability by preventing unwanted crouching behaviors without a forward command.

4) **Curriculum**: In the training environment, a total of 8192 agents are deployed in parallel to accelerate policy convergence. A curriculum learning strategy is adopted to gradually increase the task difficulty over ten progressive levels. The update policy is defined as follows:

When an agent successfully traverses more than 50% of the total terrain length, the terrain curriculum level is advanced, and additional command types are introduced.

Specifically, the command curriculum dynamically adjusts the range of control commands based on the tracking performance of key motion variables. When the reward for linear velocity tracking exceeds 90% of its maximum over an episode, the allowable range of linear velocity commands along the x-axis is expanded to encourage faster locomotion behaviors within the predefined ranges. Similarly, when the angular velocity tracking reward surpasses 70% of its maximum, the range of yaw angular velocity commands is increased to enhance turning agility and robustness.

In addition, if the commanded height approaches the limits of the current curriculum stage, the minimum and maximum height boundaries are progressively extended, thereby increasing the complexity of height regulation and improving adaptability to diverse terrains.

TABLE III
REWARDS

#	Reward	Formula	Value
1	Tracking lin vel	$\exp\left(-\frac{\ c_v - v_{\text{base}}\ ^2}{\sigma_v^2}\right)$	2
	Tracking ang vel	$\exp\left(-\frac{(c_\omega - \omega_{\text{base}})^2}{\sigma_\omega^2}\right)$	1.6
	Lin vel error	$\ c_v - v_{\text{base}}\ ^2$	-1
	Lin vel z	v_z^2	-0.05
	Ang vel xy	$\omega_x^2 + \omega_y^2$	-0.01
2	Collision	$\sum_{j \in \mathcal{X}_{\text{perality}}} \mathbb{I}(\ F_{c,j}\ > 0.1)$	-3.0
	Block interval	$R = \sum_j h_j$	-0.8
	Leg obstruction	$R = \sum_j \left(\frac{ F_{c,j} }{1000} \times b_j\right)$	-0.5
3	Nominal foot position	$\left[\frac{1}{N} \sum_{i=1}^N \exp\left(-\frac{e_{h,i}^2}{\sigma_f^2}\right)\right] \times \exp\left(-\frac{v_{\text{cmd}}^2}{\sigma_v^2}\right)$	0.8
	Tracking height	$\exp\left(-\frac{(h_{\text{base}} - c_h)^2}{\sigma_h^2}\right)$	0.35
	Base height error	$(h_{\text{base}} - c_h)^2$	-8.0
	Dof acc	$\sum_i \left(\frac{\dot{q}_{i,\text{last}} - \dot{q}_i}{\Delta t}\right)^2$	-8e-7
4	Dof pos limits	$\sum_i [\max(0, q_i^{\text{min}} - q_i) + \max(0, q_i - q_i^{\text{max}})]$	-5.0
	Torque limits	$\sum_i \max(0, \tau_i - \tau_i^{\text{limit}} \cdot r_{\text{soft}})$	-1.2
	Torque	$\sum_i \tau_i^2$	-2e-4
	Action rate	$\sum_i (a_{i,\text{last}} - a_i)^2$	-0.04
5	Keep balance	1	1.2
	Stability recovery	$\begin{cases} \max(0, -\text{sign}(\phi) \cdot \dot{\phi}) + \max(0, -\text{sign}(\theta) \cdot \dot{\theta}) \\ \text{if } \phi > 0.09 \text{ or } \theta > 0.09 \\ 0 \text{ otherwise} \end{cases}$	0.02
	No presquat	$\max(0, (c_h - h_{\text{base}}) - m) \times \mathbb{I}(v_{\text{cmd}} > v_{\text{gate}})$	-0.05

C. Active Interpolation Height Control

The preset robot height and joint inflection points were computed using MATLAB/Simscape. The corresponding visualized data were derived from the URDF file, as shown in Figs.3 and 4. By applying active linear interpolation, the joint positions associated with each target command were accurately determined. This process provides the simulated robot with a precise motion baseline, improving training efficiency. As a result, the robot can adjust its height accurately on complex terrain and maintain stability by avoiding improper leg-raising motions. Interpolation data as shown in Table.IV.

D. Velocity-triggered leg-lifting strategy

When operating on uneven terrain or encountering obstacles, the robot needs to adapt its motion to maintain effective locomotion. To address this, a velocity-triggered leg-lifting strategy is developed. It considers variations in wheel linear velocity, body velocity, and sustained forward command velocity obtained via the remote controller to trigger appropriate leg-lifting actions, which can help improve motion adaptability and overall locomotion efficiency. The leg-lifting triggered strategy is shown as

1) : When the robot receives a sustained forward command and its wheel linear velocity exceeds the body velocity, it is considered to have encountered an obstacle. In this case, an active leg-lifting action is triggered to facilitate obstacle negotiation.

2) : When both the wheels' linear velocities and body velocity approach zero while a forward command remains active, the system triggers a leg-lifting action. It helps resolve the robot stagnation issue, ensuring that the robot can continue effective locomotion even when encountering obstacles.

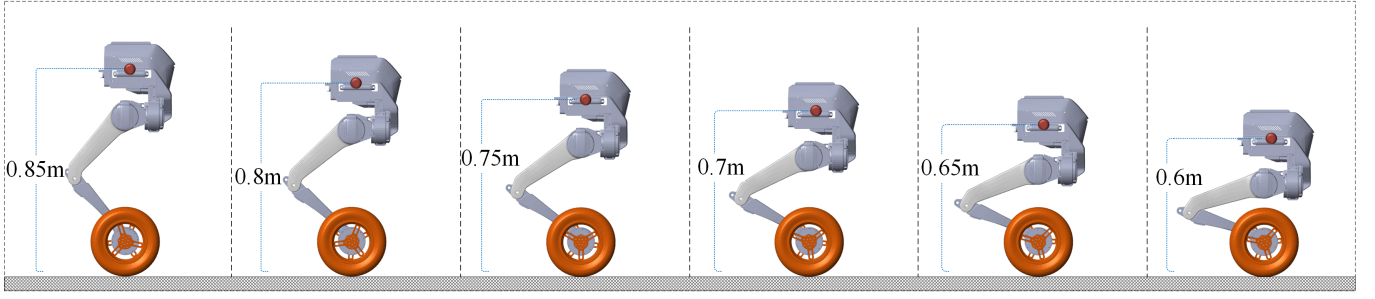


Fig. 4. Preset Height-Joint Angle Mapping Relationship.

TABLE IV
ACTIVE LINEAR INTERPOLATION DATA

Name	Data1	Data2	Data3	Data4	Data5	Data6
Height(m)	0.60	0.65	0.70	0.75	0.80	0.85
HipL(rad)	0.6083	0.5892	0.4938	0.3923	0.2823	0.1590
KneeL(rad)	1.3606	1.1784	0.9875	0.7847	0.5646	0.3181

This mechanism automatically adjusts the leg-lifting motion by continuously monitoring the contact status, obstruction intensity, and motion parameters at the robot's wheel, thereby resolving potential stalling issues when the robot encounters obstacles.

Blocked detection The system first detects the contact state of each foot. If both feet are impeded, it evaluates the obstruction intensity and assigns a weight to each foot based on the contact and tangential forces, prioritizing the lifting action on the more heavily obstructed side.

Set $\text{sat}(u) = \tanh\left(\frac{u}{v_{\max}}\right)$ as the soft upper limit for forward velocity. Set weights:

$$w_{\text{both}}^{\uparrow}, w_{\text{both}}^{\rightarrow}, w_{\text{single}}^{\uparrow}, w_{\text{single}}^{\rightarrow} > 0, \quad (1)$$

When both feet are obstructed and at nonsim heights, soft weight aggregation is employed:

$$R_{\text{nonsim}} = \sum_{i=1}^2 w_i \left(w_{\text{both}}^{\uparrow} |v_{z,i}| + w_{\text{both}}^{\rightarrow} \text{sat}(|v_{x,i}|) \right), \quad (2)$$

where, w_i is the soft weight calculated via the softmax function based on foot height. $w_{\text{both}}^{\uparrow}$ denotes the reward weight for vertical velocity when both feet are obstructed, while $w_{\text{both}}^{\rightarrow}$ represents the reward weight for forward velocity under the same condition.

When both feet are obstructed at similar heights or only one foot is obstructed, sum the obstructed feet independently:

$$R_{\text{sim}} = \sum_{i=1}^2 b_i \left(w_{\text{single}}^{\uparrow} |v_{z,i}| + w_{\text{single}}^{\rightarrow} \text{sat}(|v_{x,i}|) \right), \quad (3)$$

where, b_i is the indicator function for the obstruction state of the i -th leg, $w_{\text{single}}^{\uparrow}$ is the reward weight for vertical velocity in this scenario, and $w_{\text{single}}^{\rightarrow}$ is the reward weight for forward velocity.

When the forward stride length exceeds the limit, the robot will incur a stride penalty.

$$S_x = \max(x_1, x_2) - \min(x_1, x_2), \quad (4)$$

$$P_{\text{span}} = \lambda_{\text{span}} [S_x - S_x^{\max}]_+, \quad (5)$$

where $[u]_+ = \max(u, 0)$, S_x is the span between the two feet along the x-axis in the body coordinate system, $S_x^{\max}$ is the maximum allowable span threshold, and λ_{span} is the span penalty coefficient.

Therefore, the total reward is adjusted based on different scenarios, as shown in the following formula:

$$R = 1\{\text{nonsim}\} R_{\text{nonsim}} + 1\{\text{similar or } (b_1 + b_2 = 1)\} R_{\text{sim}} - P_{\text{span}}, \quad (6)$$

Ascending Velocity Reward: Using the weights derived from the obstruction intensity, the system provides a reward when the vertical foot velocity is positive, encouraging the robot to continue lifting that foot.

Wheel Linear Velocity Penalty: If the wheel linear velocity exceeds a predefined threshold, the system applies a penalty to suppress excessive speed, ensuring smooth leg-lifting motions and preventing instability caused by abrupt movements.

The above triggering mechanisms ensured that the robot can flexibly adjust its motion strategy according to real-time block conditions, enabling more natural and stable locomotion on complex terrains.

III. EXPERIMENTS

As shown in Fig.5, the training consists of modular terrain units within an $8 \times 8m$ area, featuring five distinct terrain configurations. Two columns are dedicated to staircase terrain, while the remaining three columns include smooth slope, rough slope, and terrain with discrete obstacles. Each terrain unit is divided into 10 progressively increasing difficulty levels: base difficulty + coefficient $\times$ difficulty. The difficulty is defined as [0.5, 0.75, 0.9].

By adding friction noise to the terrain height field, the irregularity of natural terrain is simulated. Different scales of noise are generated by adjusting parameters such as frequency, detail level, amplitude decay, and frequency growth, and then

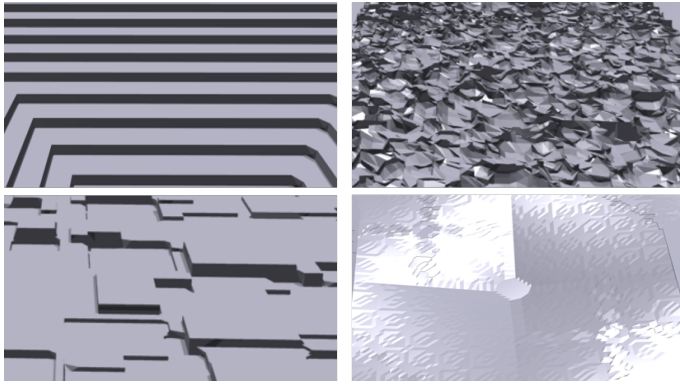


Fig. 5. First row left to right: stair terrain, discrete obstacle terrain. Second row: rough slope, smooth terrain.

overlaid onto the terrain. The generated noise enhances the terrain's variations, improving its realism.

Stair terrain: In staircase terrain design, step height increases with difficulty. Based on robot training performance, the step height is adjusted between 5cm and 10cm. The step width is fixed at 0.35, with each independent platform dimension set to 2.0 meters. This simulates stair inclination, enabling the representation of stair terrain within a single function.

Rough slope terrain: In the design of rough slope terrains, the slope adjusts with increasing difficulty. The difficulty is governed by the size of the independent platform. When the platform size falls below a predefined ratio, a slope in the opposite direction is generated, representing a smaller platform. Conversely, when the platform size exceeds the predefined ratio, a larger platform is provided.

Discrete obstacle terrain: In the design of terrains with discrete obstacles, the number of rectangles increases with difficulty. The minimum and maximum sizes of each obstacle are 1.0 and 2.0, respectively, while the platform size is maintained at 3.0 to ensure the robot can navigate obstacles of varying dimensions.

Smooth slope terrain: In the design of smooth slope terrain, the minimum and maximum elevation values define the vertical range, with a minimum of 0.0 and a maximum of 0.1. The step parameter regulates the interval of elevation changes, which is set at 0.15 to ensure discrete variations in the terrain's undulation. The downsampling factor is set to 0.2, controlling the terrain's smoothness and resolution; smaller values yield smoother terrain.

A. Simulation

In this study, we validated the proposed RL control framework using the Webots simulation platform to quantify the robot's performance in terrain adaptability and height adjustment. The validation protocol followed the training curriculum, covering four terrain types: stairs, smooth slopes, rough slopes, and discrete obstacles, with 10 difficulty levels set for each terrain type. For each terrain and difficulty level, randomized trials were conducted by varying initial poses, terrain

noise, and random seed parameters to ensure a thorough evaluation. This validated the importance of terrain information, as shown in Fig.2, which highlighted that privileged information was critical for successful learning. Fig.6 illustrates the high-fidelity simulation of the Tron1 robot in Webots.

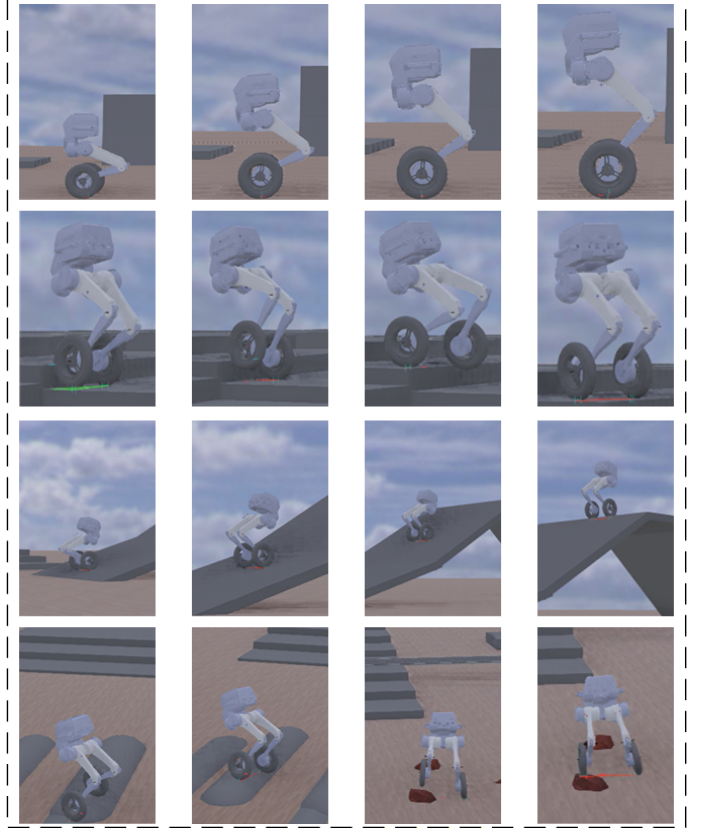


Fig. 6. Robots undergo high-assurance simulation verification in Webots for the four designed terrains. Details can be found in the supplementary video.

As shown in the second row of Fig.6, the robot navigated steps with a height of 10cm, which confirmed that the robot can traverse such steps using a velocity-triggered strategy in most cases. The third row of robots in Fig.6 demonstrated stability and a high traversal success rate on smooth slopes. It effectively adjusted height and maintained stability under moderate slopes and variable speed commands. The last row of Fig.6 illustrated that the robot faced challenges in complex terrain featuring irregular slopes and interspersed obstacles. Nevertheless, it achieved high success rates traversing such terrain, actively adjusting its posture through height control interpolation to enhance adaptability across diverse surfaces.

Active linear interpolation and velocity-triggered leg-lifting were two critical techniques in this framework. By combining height-joint angle mapping with active interpolation, the robot was able to adjust its height across diverse terrains, avoiding inappropriate posture adjustments. Simultaneously, the velocity-triggered leg-lifting strategy proactively initiated leg-lifting actions based on discrepancies between wheel speed, body velocity, and forward movement commands, ensuring

the robot could navigate obstacles smoothly.

The simulations demonstrated that the RL-based control framework effectively enabled the robot to adapt to diverse terrains while maintaining precise height regulation. The study further indicated that the robot was able to transfer behaviors learned in simulation to real-world environments, a capability subsequently validated through experiments on the Tron1.

B. Real-world Experiments

We deployed the proposed RL control framework on the 8-Degree of Freedom(8-DoF) wheeled robot Tron1. To validate the framework's performance in real-world environments, we designed diverse scenarios to test its traversal capability and posture adjustment capability.

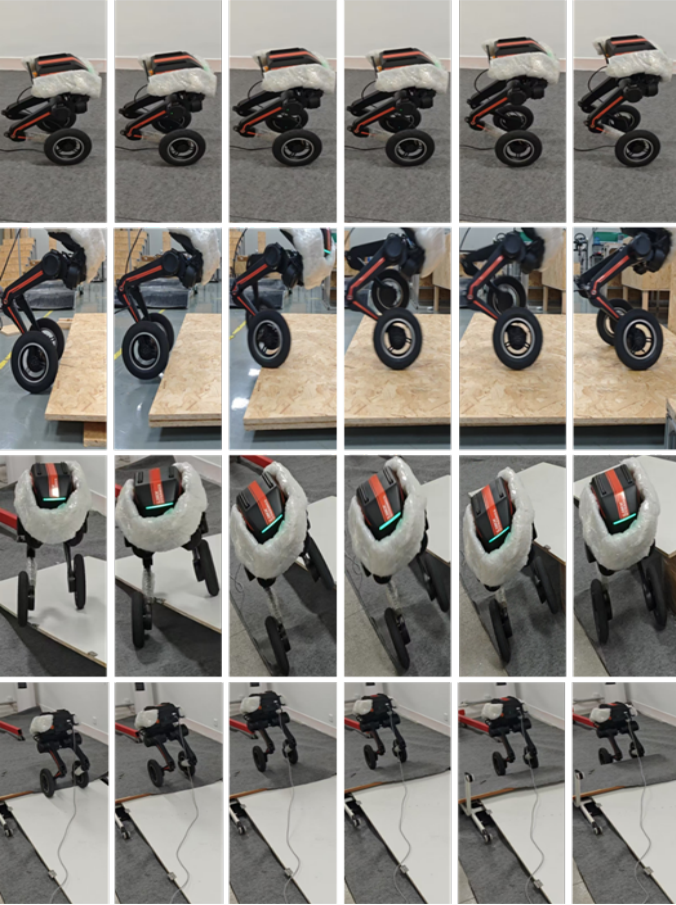


Fig. 7. Snapshots of the real world experiments. https://drive.google.com/file/d/1FsoPUZlac7x1-Ce8O_E8-E4OALCyg5IT/view?usp=sharing

The first row of Fig.7 clearly demonstrated that, while ensuring the robot's mobility, its height could be adjusted through preset height values combined with active linear interpolation. This validated the effectiveness of active interpolation height control. The second row of Fig.7 demonstrated that when encountering a 10cm step, the robot could infer the presence of an obstacle based on the tri-source velocity difference. Using a velocity-triggered strategy and weights calculated according to obstruction intensity, the system executed a leg-lifting strategy

by rewarding upward velocity when the foot's vertical velocity is positive, ensuring traversal capability.

After verifying its posture adjustment and terrain traversal ability, we suddenly issued a turning command, causing it to fall backward during the slope-climbing period of the robot. We observed that when the robot's right leg landed while the left leg remained suspended, it could still maintain balance by continuing to receive forward movement commands, as shown in the third row of Fig.7. The last row of Fig.7 shows that when traversing dynamic slopes, relative slippage occurred between the wheels and the platform. Upon reaching the slope's apex, issuing a reverse command caused the platform to dynamically oscillate, yet the robot maintained both its traversal capability and equilibrium. This demonstrates that the network exhibits exceptionally strong generalization capabilities.

The experimental results validated the effectiveness and practicality of our approach in real-world settings, confirming the success of this strategy.

IV. CONCLUSION

In summary, we trained a velocity-triggered control strategy for the B-WLR. By integrating active interpolation height control and tri-source triggered leg-lifting, this strategy enables the robot to navigate complex terrains such as stairs and slopes without relying on external sensors. Despite the encouraging results achieved in this study, several limitations remain to be addressed. Currently, the robot adjusts its height using six interpolation points, resulting in slightly lower precision. Covering all data points with the current method would require excessive preliminary effort.

Future work will replace the existing interpolation method with precise inverse kinematic calculations to provide more accurate joint angle references, thereby enhancing posture adjustment capabilities.

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REFERENCES

- [1] M. Bjelonic, V. Klemm, J. Lee, and M. Hutter, "A survey of wheeled-legged robots," in *International Conference on Climbing and Walking Robots and the Support Technologies for Mobile Machines*, 2022.
- [2] X. Liu, Y. Sun, W. Shikun, K. Cao, Q. Qi, X. Zhang, H. Shen, G. Chen, J. Xu, and A. Ji, "Development of wheel-legged biped robots: A review," *Journal of Bionic Engineering*, vol. 21, 02 2024.
- [3] J. Lee, J. Hwangbo, and M. Hutter, "Robust recovery controller for a quadrupedal robot using deep reinforcement learning," *ArXiv*, vol. abs/1901.07517, 2019.
- [4] A. Zhang, R. Zhou, T. Zhang, J. Zheng, and S. Chen, "Balance control method for bipedal wheel-legged robots based on friction feedforward linear quadratic regulator," *Sensors*, vol. 25, no. 4, 2025.
- [5] S. Liu, S. Zhou, L. Shi, H. Jing, Z. Niu, and R. Wang, "Smooth and safe stop: Fixed-time fault tolerant control for heavy legged

- robot with active identification on tolerance capability,” *ISA Transactions*, vol. 157, pp. 107–123, 2025.
- [6] Z. Xie, P. Clary, J. Dao, P. Morais, J. Hurst, and M. van de Panne, “Learning locomotion skills for cassie: Iterative design and sim-to-real,” in *Proc. Conference on Robot Learning (CORL 2019)*, 2019.
 - [7] P. M. Wensing, M. Posa, Y. Hu, A. Escande, N. Mansard, and A. D. Prete, “Optimization-based control for dynamic legged robots,” *IEEE Transactions on Robotics*, vol. 40, pp. 43–63, 2024.
 - [8] M. Bjelonic, R. Grandia, O. Harley, C. Galliard, S. Zimmermann, and M. Hutter, “Whole-body mpc and online gait sequence generation for wheeled-legged robots,” in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 8388–8395.
 - [9] S. Mason, N. Rotella, S. Schaal, and L. Righetti, “An mpc walking framework with external contact forces,” in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 1785–1790.
 - [10] Z. Pan, B. Li, S. Zhou, S. Liu, Z. Niu, and R. Wang, “Distributed mpc-based posture control for knee-wheeled wheel-legged robots with multi-actuation,” *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, vol. 238, no. 14, pp. 4458–4471, 2024.
 - [11] P. Agarwal and K. et al., *State Estimation for Legged Robots: Consistent Fusion of Leg Kinematics and IMU*. MIT Press, 2013, pp. 17–24.
 - [12] S. Liu, S. Zhou, Z. Pan, Z. Niu, and R. Wang, “Load awareness: Sensorless body payload sensing and localization for heavy quadruped robot,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 759–766.
 - [13] M. V. Minniti, R. Grandia, F. Farshidian, and M. Hutter, “Adaptive clf-mpc with application to quadrupedal robots,” *IEEE Robotics and Automation Letters*, vol. 7, no. 1, pp. 565–572, 2022.
 - [14] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, “Learning agile and dynamic motor skills for legged robots,” *Science Robotics*, vol. 4, no. 26, Jan. 2019.
 - [15] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, “Learning to walk in minutes using massively parallel deep reinforcement learning,” 2022.
 - [16] N. Rudin, D. Hoeller, M. Bjelonic, and M. Hutter, “Advanced skills by learning locomotion and local navigation end-to-end,” 2022.
 - [17] J. Lee, M. Bjelonic, and M. Hutter, *Control of Wheeled-Legged Quadrupeds Using Deep Reinforcement Learning*. MIT Press, 01 2023, pp. 119–127.
 - [18] S. Chamorro, V. Klemm, M. de la Iglesia Valls, C. Pal, and R. Siegwart, “Reinforcement learning for blind stair climbing with legged and wheeled-legged robots,” 2024.
 - [19] R. Li, H. Wang, Q. Li, Z. Han, Y. Chu, L. Ye, W. Xie, and W. Liao, “Ctbc: Contact-triggered blind climbing for wheeled bipedal robots with instruction learning and reinforcement learning,” 2025.
 - [20] J. Lee, M. Bjelonic, and R. et al., “Learning robust autonomous navigation and locomotion for wheeled-legged robots,” *Science Robotics*, vol. 9, no. 89, Apr. 2024.
 - [21] V. Makovychuk, “Isaac gym: High performance gpu-based physics simulation for robot learning,” 2021.
 - [22] O. Michel, “Webotstm: Professional mobile robot simulation,” 2004.
 - [23] S. Zhou, S. Liu, Z. Lin, Z. Niu, Z. Pan, and R. Wang, “Cascade trajectory optimization with phase duration adaption and control for wheel-legged robots overcoming high obstacles,” in *2023 International Conference on Advanced Robotics and Mechatronics (ICARM)*, 2023, pp. 832–839.